Master's in Informatics and Computing Engineering

Department of Informatics Engineering

Dissertation Preparation

Topics on ethics and ethical AI usage

by Alexandre Valle and Daniel Mendes

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U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

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Agenda

Research ethics and integrity

What the profession expects of you and why

AI in academic writing

Actions allowed, declaration requirements, and what remains yours

Morality and ethics

Greek ***ēthos*** as “character”, “habit” or “custom”.

Latin ***moralis*** as pertaining to “custom”, “habit”, or “social conduct”

Dimension	**Morality**	**Ethics**
Level	First-order: a phenomenon through which communities and individuals evaluate actions, character, and social arrangements (right/wrong)	Second-order: philosophical study of moral phenomena
Nature	Lived, often intuitive and culturally transmitted	Analytical, argued, systematic
Source of authority	Culture, upbringing, conscience, evolved psychology	Reasons, principles, theories
Scope	Personal and communal	Universal in aspiration, or role-specific (professional codes)
Study	Moral psychology, anthropology	Philosophy
Typical output	“Lying is wrong”	“What grounds that claim, and when might it fail?”

Ethics is the systematic, reflective study of morality: it analyses, justifies, criticises and compares moral norms and asks what makes actions right or wrong.

Why computing has a code of ethics?

Some practices (medicine, engineering, law) have codes because their practitioners can cause harm.

A code is not a rulebook with penalties attached

It is a statement of what the profession expects, written by the professionals

It gives you language for objections you would otherwise struggle to raise

The ACM Code of Ethics is one of the references for computing.

Your dissertation is an academic and a professional work. This Code also applies.

The ACM Code of Ethics and Professional Conduct

1. General ethical principles (1.1 – 1.7)

What every computing professional owes

2. Professional responsibilities (2.1 – 2.9)

How those principles apply to doing the work

3. Professional leadership principles (3.1 – 3.7)

For those who direct the work of others

4. Compliance with the code (4.1 – 4.2)

Upholding it and the consequences of not doing so

You should consult the full code at **https://www.acm.org/code-of-ethics**

1.3 - Be honest and trustworthy

Honesty is a prerequisite for trust:

Do not misrepresent your results, your methods, or your contributions

Do not misrepresent the work of others

Disclose anything that bears on how your work should be read

Be honest about limitations and about what you do not know

Omitting an inconvenient result is a form of misrepresentation, even when every sentence you write is true.

1.5 - Respect the work required to produce new ideas

The imperative that governs most of what you do in a dissertation.

Credit the creators of ideas, inventions, work and artefacts

Respect copyrights, patents, licences and other protections

Do not claim private ownership of what was shared as a public resource

Citation is the mechanism by which this profession pays its debts.

1.6 and 1.7 - Privacy and confidentiality

Relevant if your dissertation involves users, user data, or a company.

Privacy

- Collect only what you need
- Inform participants and collect their consent
- Protect data you hold (and destroy when no longer needed)
- Anonymise before analysis, not before publication

Confidentiality

- Honour non-disclosure terms you agreed to
- Agree early what can be published - not when you are writing it up

Remember that a thesis is ALWAYS publicly defended (president, supervisor and examiner require full access)

You can request that the work not be published after the dissertation for up to 3 years.

2.1 and 2.5 - Quality and thorough evaluation

2.1 Strive to achieve high quality in both the processes and products of professional work.

2.5 Give comprehensive and thorough evaluations, including analysis of possible risks.

Should be an ethical obligation, not just a professional one.

Evaluate your own artefact as critically as you would someone else's

Report what failed, not only what worked

When applicable, consider who could be harmed by what you built and say so

A thorough evaluation that produces an inconvenient result is better work than a selective one that does not.

Plagiarism

From Latin “*plagium*”: to kidnap, to seize, to appropriate

Using someone else's words, ideas, structure or results without proper attribution.

Verbatim text without quotation marks, even with a citation

An idea presented as your own, even in your own words

A figure, table or dataset reused without credit

Code copied without attribution or licence compliance

Patchwriting

The failure most students fall into without meaning to.

Taking a source sentence and changing some words, reordering clauses, swapping synonyms - while keeping its structure.

It happens when you draft with the source open beside you

It happens when you “save segments to cite later“

It happens most when you do not fully understand the source

Paraphrase or patchwrite example

Prefer

Source: "Adaptive scheduling reduces tail latency by reallocating idle cores during traffic bursts."

Yours: Costa et al. attribute the latency gain to reallocation during bursts - a mechanism that assumes idle capacity is available [12].

→ restructured and extends the source with your own observation

Avoid

Source: "Adaptive scheduling reduces tail latency by reallocating idle cores during traffic bursts."

*Yours: Adaptive scheduling **lowers** tail latency **through** the reallocation of **unused cores** during **bursts of traffic** [12].*

→ same structure order, synonyms substituted

→ this is patchwriting, and the citation does not fix it

Self-plagiarism and text recycling

Reusing your own previous writing without being transparent about it.

You can plagiarise yourself.

The academic record is distorted either way, and the reader is misled about what is new.

Where it may hurt in a dissertation

Text from your own earlier reports, papers or coursework

What to do

Cite your own prior work whenever text or reasoning is reused

Declare the overlap, and say what is new

Check what rights you retained - the publisher may own the text you wrote

Figures, data, and permission

Citation is not permission.

To ethically use another author's figure in a scientific publication, you must:

- obtain copyright permission (if required)
- cite the original source in the caption
- explicitly state whether the figure is reproduced, adapted, or redrawn.

Datasets and code

Check the licence. Open access is not the same as public domain.

When is plagiarism found

Detection is not the risk students imagine it to be.

The real risk is slower and worse.

Before submission - your supervisor asks you to rewrite. Time lost.

At the defense - a jury member recognises a passage. Trust in the whole document degrades.

After graduation - a thesis is a permanent, public, searchable document.

The asymmetry is considerable: the shortcut saves hours, but the consequence lasts a career.

AI in academic writing

What is permitted, what must be declared, and what remains yours

The position: transparency, not prohibition

Generative AI may be used. Its use must be disclosed.

AI tools may not be listed as authors - they cannot take responsibility

Use to create content is permitted, but must be fully disclosed in the work

Basic aids - spelling and grammar checkers - are exempt

If uncertain whether something needs disclosing, disclose it

You remain responsible for every word, every claim and every citation, whatever produced it.

What these tools do well, and badly

Reasonable uses

Explain a concept you are stuck on, at several levels

Aid in structuring something you will write yourself

Criticise your draft, or rehearse for the defence

Unreliable

Anything factual you cannot check

References, which are produced fluently and often do not exist, or do exist but you did not check its contents.

Summaries of papers

Perform the critical analysis for you!

Verification

Never cite a work you have not read and verified the original content.

This rule predates AI.

Check every reference against the publisher's record

Check every number, every claim, every piece of code

Check that a cited work says what you say it says

If you cannot verify it, you cannot use it.

Fabricated references

The failure is specific, predictable, and easy to catch.

Plausible authors, a real-sounding venue, a well-formed DOI that resolves to nothing - or to a different paper entirely.

Mixed in among real references, so a spot check passes

Often on exactly the claim you most wanted supported

One invented reference discredits every other one in your list.

Prompts that work against you

"Write this chapter for me, with academic references"

replaces your intellectual work and invites fabricated citations

"Do a complete literature review on X"

gives a false sense of mastery, built on sources you never read

"Solve this completely and explain afterwards"

skips the initial effort, which is where the learning happens

"Summarise this paper and tell me what to say about it"

replaces your critical reading with someone else's

Prompts that (may) help

"Explain this concept at three levels, and say where the explanation breaks down"

"Based on these three sources I have read, suggest comparisons and gaps I should explore. Do not invent references."

"Give me an initial hint for this problem"

"Act as a sceptical reviewer: what must I confirm in the original before submitting?"

"Ask me six hard questions about this chapter. Do not give answers."

The AI answers may be complementary perspectives, but your critical thinking is still mandatory!

When using AI keep a usage trail

Record as you go!

Reconstructing it afterwards is unreliable and looks evasive.

- Which tool and which version
- What you used it for
- Which parts of the document are affected
- What you verified yourself and how you verified it

A short note in your repository is enough. It costs a line a day.

You cannot accurately fill out the declaration of Honor if you did not record the AI usage.

Declaring use of AI

Disclosure goes in the Declaration of Honour (FEUP M.EIC template).

Name tools and versions

State the purposes: language, structure, code, examples,

Identify the affected parts of the work

State the verifications you performed

Affirm what in the document is strictly yours

A disclosure is a statement of facts, not a confession or a disclaimer.

Disclosure statement examples

Prefer

Generative AI tools (Claude, Opus 5, high) were used to support linguistic revision and to vary illustrative examples. The selection of sources, the critical analysis, the conclusions and the final text were produced and verified by the author, who assumes full responsibility for this document.

→ names purpose and scope

→ locates responsibility unambiguously

No generative AI tool was used in any part of the work or in any part of the dissertation writing process. The Author assumes full responsibility for this document.

→ objective: no use at all

→ full responsibility

Avoid

AI was used in the preparation of this document.

→ declares nothing useful: which tool, for what, where?

No statement at all, where tools were used.

→ FORBIDDEN!

Defending AI usage questions

Expect to be asked.

Prepare for it as you would any other part of the defence.

Where did you use it, and why there?

What did you verify yourself, and how?

Which parts are entirely your own?

Explain this passage of the document.

The last one is a real test. If you cannot explain your own text, it is not yours.

What remains your full responsibility

All the key parts of the argument!

The problem you identified and the question(s) you chose to ask.

The judgement about what matters in the literature.

All parts of the critical analysis

The design decisions, and the reasons for them.

The interpretation of what the evidence means.

The responsibility for every claim in the document.

None of this can be delegated.

Final thoughts

These tools exist (and new tools are emerging):

Discovery and citation mapping (Connected Papers, ResearchRabbit, Litmaps, Scite)

Screening and systematic review management (Covidence, Rayyan, DistillerSR)

AI research assistants exist (Elicit, Consensus, SciSpace, Undermind, Scite, and Paperguide)

Etc., etc.

(They keep getting better every day…)

Some may bring added value. They don’t replace you in all the critical parts of the work! And you need to disclose their use.

Dissertation Preparation

Questions?